

Research Synthesis — Pre-Build Literature & Data Review

Written 2026-08-29, concluding the research/doc-hunting phase before implementation begins. This is not a from-scratch academic paper — it's a synthesis of real, searched sources plus this project's own design decisions, structured like a short literature review so the reasoning behind `docs/algorithms/*.md` is traceable to something more solid than "seemed reasonable." Every claim below was checked with a live search or fetch on 2026-08-29 unless marked otherwise; sources are linked inline rather than footnoted, per this project's own copyright/citation discipline.

1. The core problem is "entity resolution," a 55-year-old research area

What this challenge calls "matching a company profile to verified public evidence" is, in the data-management literature, **entity resolution** (also called record linkage or deduplication). The foundational result is Fellegi & Sunter's 1969 paper, *A Theory for Record Linkage* (published in the *Journal of the American Statistical Association*), which framed matching as a decision problem with three outcomes — link, possible-link, non-link — and showed the likelihood-ratio test that minimizes false links/false non-links at chosen error rates. That 1969 framework is still the basis of most production record-linkage tooling today (surveyed in Christen's book *Data Matching* and Winkler's 2006 survey; see also the arXiv survey *"(Almost) All of Entity Resolution"*, <https://arxiv.org/pdf/2008.04443>).

Why this matters for the build: the current plan

(`docs/algorithms/match_algorithm.md`) uses a single RapidFuzz similarity score plus a two-threshold rule (auto-accept above 92, auto-reject below 75, corroborating-signal-required in between). That's a reasonable, fast starting point, but it's a simplified special case of Fellegi-Sunter — it only looks at one field (name) instead of combining multiple weak signals (name similarity + address match + org number found in page text + industry code match) into a single

probabilistic score.

Concrete option worth evaluating: [Splink](https://github.com/moj-analytical-services/splink)

is a maintained, open-source Python library (used by the UK Ministry of Justice and cited in an ONS case study on 2021 Census linkage) that implements Fellegi-Sunter-style probabilistic matching across multiple fields, with unsupervised parameter estimation via expectation-maximization — meaning it doesn't require hand-labeled training pairs to get started. If dev-slice testing shows the simple RapidFuzz threshold rule producing too many borderline rejects/accepts, Splink (or a hand-rolled multi-field weighted score following the same principle) is the natural upgrade path, not a rewrite — see "Recommendations" below.

2. Ground truth exists and is under-used in the original plan

The single most important finding of this research phase: Norway's official company registry (Brønnøysundregisteret's Enhetsregisteret) has a free, public, no-auth API, and it directly answers several fields the original crawl-first plan assumed had to be *discovered*:

- `hjemmeside` — the registered website (tier-1 official-site candidate, no matching/confidence-scoring needed)
- `antallAnsatte` — employee count (directly answers `headcount_band`, no LinkedIn inference needed)
- `overordnetEnhet` on the `/underenheter` endpoint — structured parent/subsidiary links (directly answers the brief's called-out "don't confuse a brand, subsidiary, or parent" failure mode)
- A nightly bulk-download endpoint for the entire register, and an incremental-updates endpoint — both map almost exactly onto this project's own "resolve once, refresh via diff" architecture

Full verified details: `docs/algorithms/registry_api.md` (updated this session from live API documentation, replacing an earlier version written from unverified training-time knowledge — see that file's changelog note for what specifically changed).

Implication: coverage and precision on the *easy* majority of the 411,160 companies should come almost entirely from this API and the bulk download, essentially for free (no crawl, no matching, no confidence scoring needed for those fields). Crawl/match/extract effort should concentrate on the harder fields the registry doesn't carry — LinkedIn presence, hiring signal, leadership names, dated public activity — which is exactly where the brief's "not just the easy companies or the registry record" language says the real scoring differentiation lives.

3. Structured-data extraction has a real, measurable ceiling — plan around it, not against it

Search results (Web Data Commons project, University of Mannheim, 2025 analysis of the Common Crawl corpus) indicate that roughly 44% of web pages now carry some Schema.org structured-data markup, up from about 7.5% in 2013. That's a real, growing majority-adjacent figure — but it means a JSON-LD-first extraction strategy (docs/algorithms/extract_algorithm.md) will still come up empty on a meaningful minority of company sites, especially smaller Norwegian businesses less likely to have invested in SEO tooling that adds Organization schema automatically.

Implication for coverage strategy: the extraction fallback chain (JSON-LD → OpenGraph/microdata → Trafilatura text) isn't just a nice progressive-enhancement pattern, it's structurally necessary — treating Trafilatura text extraction as a "sometimes," not "rare," path is more realistic than the original doc implied. Budget accordingly: don't assume most sites will short-circuit at step 1.

4. Open questions this research did NOT resolve (flag before building)

- **LinkedIn and job-board scraping legality/ToS boundaries** — not independently verified this session; docs/source_policy.md still needs the real Builderr source-policy document pasted in before any

src/crawl/ spider targets these sources. This is the single highest-risk unresolved item, since it directly touches the "documented source rights" hard gate.

- **Whether the br reg PyPI client is actively maintained** — found it exists (Otovo AS, Apache 2.0) but did not check recent commit history or issue activity. Verify before depending on it versus hand-rolling the ~4 endpoints actually needed.
- **Splink's practical overhead** for a challenge with a 45-minute per-batch wall-clock budget — it's designed for large-scale offline linkage jobs, not necessarily a tight per-batch runtime budget. Needs a small benchmark on the dev slice before committing to it over the simpler RapidFuzz approach.

5. Recommendations (concrete, prioritized)

1. **Rebuild src/resolve/** around the bulk-download + live-lookup split** described in the updated docs/algorithms/registry_api.md, not per-company live calls for the full universe — this is now the single highest-leverage architecture change from this research pass.
2. **Extract hjemmeside, antallAnsatte, and overordnetEnhet** directly from registry data** as FOUND claims with the registry itself as provenance, before any crawling starts for those specific fields — free coverage and precision points.
3. Keep the RapidFuzz two-threshold matcher as the Phase-1 implementation (it's simple, fast, and already tested — see src/match/normalize.py), but log enough per-decision detail (which signals contributed) to make a later move to a proper multi-field Fellegi-Sunter-style score (Splink or hand-rolled) a data-driven decision, not a guess, once real dev-slice results exist.
4. **Don't under-resource Trafilaturation-based text extraction** — treat it as a first-class path, not a rare fallback, given the ~44% structured-data ceiling.
5. Resolve the source-policy and br reg-client maintenance questions

above before Phase 2 (crawl) begins — both are cheap to check and block real decisions downstream.

Sources consulted this session

- Brønnøysundregisteret Enhetsregisteret API docs — <https://data.brreg.no/enhetsregisteret/api/docs/index.html>
- Fellegi, I.P. & Sunter, A.B. (1969), *A Theory for Record Linkage*, cited via <https://arxiv.org/pdf/2008.04443> and <https://arxiv.org/pdf/2104.09677>
- Splink (probabilistic record linkage library) — <https://github.com/moj-analytical-services/splink>
- Web Data Commons / Schema.org adoption statistics — via search results citing the University of Mannheim's Common Crawl analysis
- brreg Python client — <https://pypi.org/project/brreg/1.0.0a1>